

Experiment No 1

Port Scanning Tools

Aim

To study and perform different types of **port scanning techniques using Nmap in Kali Linux**, such as TCP SYN, TCP Connect, UDP, and TCP ACK scans, to identify open, closed, and filtered ports on a target system.

Algorithm

1. Start Kali Linux and open the **Nmap** tool.
2. Select **Applications** → **Information Gathering** → **Nmap**.
3. Identify a target system for scanning, preferably your own machine or an authorized lab system.
4. Perform a **TCP SYN scan** using:

```
nmap -sS <target-IP>
```
5. Perform a **TCP Connect scan** using:

```
nmap -sT <target-IP>
```
6. Perform a **UDP scan** using:

```
nmap -sU <target-IP>
```
7. Perform a **TCP ACK scan** using:

```
nmap -sA <target-IP>
```
8. Observe the scan results and note the status of the ports.
9. Compare the results obtained from the different scanning techniques.
10. Stop the scan and record the observations.

Output

```
labex:project/ $ sudo nmap -sS 127.0.0.1
Starting Nmap 7.80 ( https://nmap.org ) at 2026-10-01 12:46 CST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
Not shown: 997 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh
3000/tcp   open  ppp
3001/tcp   open  nessus

Nmap done: 1 IP address (1 host up) scanned in 0.06 seconds
labex:project/ $ sudo nmap -sT 127.0.0.1
Starting Nmap 7.80 ( https://nmap.org ) at 2026-10-01 12:48 CST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000056s latency).
Not shown: 997 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh
3000/tcp   open  ppp
3001/tcp   open  nessus

Nmap done: 1 IP address (1 host up) scanned in 0.03 seconds
labex:project/ $ sudo nmap -sU 127.0.0.1
Starting Nmap 7.80 ( https://nmap.org ) at 2026-10-01 12:48 CST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
All 1000 scanned ports on localhost (127.0.0.1) are closed

Nmap done: 1 IP address (1 host up) scanned in 0.05 seconds
labex:project/ $ sudo nmap -sA 127.0.0.1
Starting Nmap 7.80 ( https://nmap.org ) at 2026-10-01 12:49 CST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
All 1000 scanned ports on localhost (127.0.0.1) are unfiltered

Nmap done: 1 IP address (1 host up) scanned in 0.05 seconds
```

Result

Thus, different **Nmap port scanning techniques** such as **TCP SYN**, **TCP Connect**, **UDP**, and **TCP ACK scans** were successfully performed in Kali Linux. The scans helped identify the **open**, **closed**, and **filtered port states** of the authorized target system.